

Exercise 5: Advanced SQL - NULL Functions

--- Q1: REPLACING NULL VALUE

```
SELECT employee_id,  
       name,  
       IFNULL(salary, 0) AS salary  
FROM Employees;
```

employee_id	name	salary
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

--- Q2: REPLACING NULL VALUE

```
SELECT employee_id,  
       name,  
       IFNULL(department, 'Not Assigned') AS department  
FROM Employees;
```

employee_id	name	department
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

--- Q3: FINDING NULL VALUES

```
SELECT order_id,  
       customer_id  
FROM Orders  
WHERE customer_id IS NULL
```


order_id	customer_id
103	NULL

--- Q4: REPLACING NULL VALUE

```
SELECT order_id,
       customer_id,
       COALESCE(delivery_date, 'Pending') AS delivery_status
FROM Orders;
```

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

--- Q5: REPLACING NULL VALUE

```
SELECT student_id,
       name,
       IFNULL(grade, 0) AS final_grade
FROM Students;
```

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

--- Q6: COUNT NULL VALUES

```
SELECT COUNT(grade) AS not_graded_count
FROM Students
WHERE grade IS NULL;
```


not_graded_count
1

---Q7: CALCULATE DISCOUNTED PRICE + REPLACE NULL

```
SELECT product_id,
       name,
       price - COALESCE(discount, 0) AS final_price
FROM Products
```

product_id	name	final_price
S01	Keyboard	25
S02	Mouse	10
S03	Monitor	100

---Q8: COUNT NULLS

```
SELECT COUNT(emails) AS missing_email_count
FROM Customers
WHERE email IS NULL
```

missing_email_count
2

---Q9: REPLACE NULL

```
SELECT customer_id,
       name,
       IFNULL(email, 'No Email') AS email_display
```

customer_id	name	email_display
1	Linda	No Email
2	Joseph	joseph@gmail.com
3	Nia	No Email

-- Q10: REPLACE NULL VALUES

```
SELECT payment_id,
       IFNULL(method, 'Unknown') AS method_display,
       status
FROM Payments;
```

payment_id	method_display	status
301	Credit	NULL
302	Pay Pal	Success
303	Unknown	Failed

-- Q11: REPLACE NULL VALUE

```
SELECT item_id,
       name,
       IFNULL(quantity, 0) AS quantity_checked
FROM Inventory;
```

item_id	name	quantity_checked
1	Pen	0
2	Notebook	150
3	Erasen	0

--- Q12: USE COALESCE

```
SELECT emp_id
       COALESCE(bonus, commission) AS first_available_reward
FROM Employees_Extra;
```

emp-id	first_available_reward
1	300
2	100
3	NULL

---Q13: COUNT NULL VALUES

```
SELECT COUNT(room) AS no-room-count
FROM Classes
WHERE room IS NULL;
```

no-room-count
2

---Q14: REPLACE NULL VALUES

```
SELECT account_id,
       IFNULL(account_type, 'Unknown') AS type-display,
       IFNULL(balance, 0) AS balance-checked
FROM Bank-Accounts;
```

account_id	type-display	balance-checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

---Q15: REPLACE NULL VALUE

```
SELECT student_id,
       date,
       IFNULL(status, 'Not Marked') AS attendance-status
FROM Attendance;
```

student_id	date	attendance-status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

-- Q16: REPLACE NULL

```
SELECT project_id,  
       title,  
       IFNULL(start_date, 'TBD') AS start_display  
FROM Projects;
```

project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

--- Q17: REPLACING NULL VALUES

```
SELECT review_id,  
       product_id,  
       IFNULL(comment, 'No Comment') AS comment_display  
       IFNULL(rating, 0) AS rating_display  
FROM Reviews;
```

review_id	product_id	comment_display	rating_display
1	501	Great Product	4
2	502	No Comment	0
3	503	Works fine	3

-- Q18: USE COALESCE

```
SELECT supplier_id,  
       name,  
       COALESCE(phone, alt_phone, 'No Contact') AS contact_number  
FROM Suppliers;
```


supplier_id	name	contact_number
1	Global Goods	123456789 987654321 No Contact
2	Best Supplies	
3	Value Source	No Contact.

---Q19: REPLACE NULL VALUES

```
SELECT user_id,
       IFNULL(theme, 'Light') AS theme_set,
       IFNULL(language, 'English') AS language_set,
       IFNULL(timezone, 'UTC') AS timezone_set
FROM user_settings;
```

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

---Q20: REPLACE NULL VALUES

```
SELECT record_id,
       machine_id,
       IFNULL(issue, 'Unknown Issue') AS issue_log,
       IFNULL(technician, 'Not Assigned') AS technician_name
FROM Maintenance;
```

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex